Hanchen (Howard) Xiao

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RESEARCH PROFILE

Second-year Stanford Electrical Engineering PhD student advised by Prof. Gordon Wetzstein, researching image and video generation models. Research interests span the fundamentals of diffusion and flow-based generative models, foundation models, long-context and causal video generation, and world models for physical AI.

EDUCATION

Stanford University, PhD in Electrical Engineering *Sep. 2025–Present*
Advisor: Prof. Gordon Wetzstein | Anticipated graduation: June 2029 | GPA: 4.00/4.00

University of Toronto, Honours BSc in Computer Science and Mathematics *Sep. 2020–Jun. 2025*
Computer Science Specialist and Mathematics Specialist | GPA: 3.86/4.00

SELECTED RESEARCH

Spectral Progressive Diffusion for Efficient Image and Video Generation — Preprint *2026*
Howard Xiao, Brian Chao, Lior Yariv, Gordon Wetzstein

- Developed a general framework that progressively increases spatial resolution along the diffusion denoising trajectories using spectral representations and power-spectrum-derived resolution schedules. The method supports both training-free inference and LoRA fine-tuning without architecture modifications, accelerating inference while preserving quality.
- Evaluated latent-space image, pixel-space image, and latent-space video generation models including FLUX.1-dev, Z-Image, PixelGen, and WAN 2.1, achieving up to 7.09× wall-clock speedup on FLUX and 2.54× on WAN 2.1 while maintaining generation quality.
  Project | **Paper** | **Code** | **Demo**

Policy-based Foveated Imaging and Perception — SIGGRAPH 2026 *2026*
Howard Xiao, Jan Ackermann, Boyang Deng, Gordon Wetzstein

- Designed and evaluated a policy-based foveated imaging system that adaptively allocates sensing bandwidth for downstream perception tasks using ultra-high-resolution sensors.
- Validated the method in simulation and on a 200-megapixel sensor prototype, preserving task performance across object tracking, text recognition, and robotic manipulation tasks while using less than 1/8 of the full-resolution pixel bandwidth.
  Project | **Paper**

Foveated Diffusion: Efficient Spatially Adaptive Image and Video Generation — Preprint *2026*
Brian Chao*, Lior Yariv*, **Howard Xiao**, Gordon Wetzstein * *Equal contribution*

- Introduced mixed-resolution image and video generation for pretrained diffusion transformers, preserving high-resolution tokens in foveal regions while significantly reducing peripheral computation.
- Applied LoRA post-training to image and video generation models including FLUX.2 and WAN 2.1, achieving 2× wall-clock speedup for image generation and 4× for video generation while maintaining generation quality.
  Project | **Paper** | **Code**

TECHNICAL SKILLS

Research: Generative AI; Diffusion Models; Flow Matching; Image and Video Generation; Efficient Generative Models; Computer Vision; Long-Context Modeling; Causal Video Generation; World Models; Vision-Language-Action Models; World Action Models; Physical AI
Programming & Frameworks: Python; PyTorch;
ML Systems: Distributed and Multi-GPU Training; Model Fine-Tuning; Efficient Attention; Inference Optimization

AWARDS

Stanford Graduate Fellowship in Science and Engineering (SGF) *Sep. 2025–Sep. 2028*
Undergraduate Student Research Award (USRA) *May–Aug. 2024*
Supervisors: Prof. Kyros Kutulakos and Prof. David Lindell | Ultra-wideband Single-photon 3D Imaging

ADDITIONAL PUBLICATIONS

[1] Sotiris Nousias*, Mian Wei*, **Howard Xiao**, Maxx Wu, Shahmeer Athar, Kevin J. Wang, Anagh Malik, David A. Barmherzig, David B. Lindell, and Kyros Kutulakos. "Opportunistic Single-Photon Time of Flight." **CVPR**, 2025 (**Oral presentation**).
[2] Anton Izosimov, Boris Khesin, and **Howard Xiao**. "Virasoro Extensions for Diffeomorphisms with Breaks." Preprint, 2026.
* *Equal contribution*

INDUSTRY EXPERIENCE

Software Developer Intern, Bell Canada, Mississauga, ON *May 2023–Aug. 2024*

- Built internal document-retrieval and code-generation tools with Python, Ruby, SQL, and LangChain, and fine-tuned open-source language models during a 16-month internship.

TEACHING

Advanced Linear Algebra, University of Toronto, Teaching Assistant *May 2023–May 2024*